

Impact of XPM in WDM Fiber-Optic Transmission System in Presence of Second Order GVD

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Abstract— In wavelength division multiplexing (WDM) optical transmission system, crosstalk due to cross-phase modulation (XPM) is one of the most significant nonlinear effect. On the other hand in a first order group velocity dispersion (GVD) compensated system, second order GVD also restricts bit rate. This paper presents an analytical treatment of XPM effect in presence of second order GVD in WDM fiber-optic transmission system. It is found that in long haul high bit rate system, second order GVD plays a critical role in limiting performance in a WDM system. Results show that XPM crosstalk penalty in presence of second order GVD is 43 dB more for dispersion shifted fiber than standard single mode fiber at 10 GHz modulation frequency and 0.8 nm channel spacing.

Keywords— Cross-phase modulation, group velocity dispersion, wavelength division multiplexing and modulation frequency.

I. INTRODUCTION

Optical transmission is a preferred medium for long distance, high bandwidth communication system running at speed in the range of gigabit per second or higher. The process towards ever increasing speed encounters an obstacle in the form of the group velocity dispersion (GVD) in the optical fiber that restricts bit rates [1-2]. On the other hand, the progress towards longer lengths of transmission has led to increasing input power and at higher power; nonlinear effects are introduced in the optical fiber that in turn also limits the transmission distance of optical signal [3-4]. Cross-phase modulation (XPM) is one of the most significant nonlinear effects that impact the system performance significantly [5-6]. As a result, the combined effects of GVD and XPM may cause further deterioration of transmission performance in a WDM system. But first order GVD can be totally compensated using different technologies [7] and then the effect of dispersion slope which arises from second order GVD plays a critical role in limiting the system performance in long haul high bit rate transmission system.

Hui *et al.* (1998) evaluated spectral characteristics of XPM in multi-span WDM systems both experimentally and theoretically [8]. They found that the crosstalk level is dependent on optical channel spacing and fiber dispersion. Interference between XPM crosstalk generated in different fiber spans creates strong ripples in the frequency response. Thus, it is essential to study the impact of XPM and second order GVD to predict the performance limitation or system outage accurately due to these effects in the WDM transmission system. In this paper, we have studied the impact of XPM in presence of second order GVD in multi-channel multi-span WDM fiber-optic transmission system for

3-different types of fiber such as - standard single mode fiber (SSMF), dispersion shifted fiber (DSF) and large effective area fiber (LEAF).

II. SYSTEM MODEL

The block diagram of a multi span multi channel WDM system with EDFA in cascade used for theoretical analysis is shown in Fig. 1. The N channels are multiplexed by WDM MUX and the composite signal is transmitted through an optical fiber. To describe the effect of XPM, we assume that one channel is probe which is operated at 1559 nm wavelength is a low power continuous wave and the remaining channels are pump. The in-line EDFA's are used in cascade to compensate the fiber losses. Finally, the composite signal is demultiplexed at WDM DEMUX and from the modulated carrier, original signal is recovered through IM-DD method.

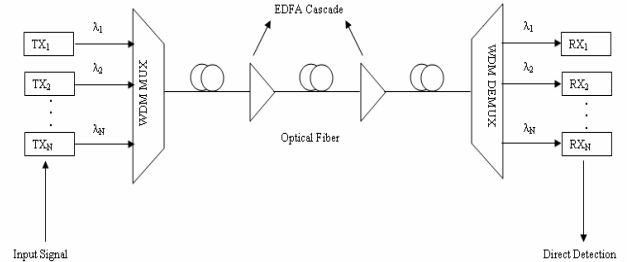


Fig. 1: Block diagram of a WDM system with EDFA in cascade.

III. PUMP-PROBE PROPAGATION EQUATION AND CROSSTALK

Using the slowly-varying envelope approximation, the propagation of the channels along the transmission medium is governed by the nonlinear Schrödinger equation (NLS). The general form of NLS considering attenuation factor, the first- and second-order GVD as well as the SPM and XPM effects for an arbitrary channel k is given by,

$$\frac{\partial A_k}{\partial z} + \frac{\alpha_k}{2} A_k + \frac{1}{v_{gk}} \cdot \frac{\partial A_k}{\partial t} + \frac{i}{2} \beta_{2k} \frac{\partial^2 A_k}{\partial t^2} - \frac{1}{6} \beta_{3k} \frac{\partial^3 A_k}{\partial t^3} = i \gamma_k \left[|A_k|^2 + 2 \sum_{j=1, j \neq k}^N |A_j|^2 \right] A_k \quad (1)$$

where k is the current channel under consideration, N is the total number of channels, α_k represents the attenuation parameter, β_{2k} and β_{3k} are the first -and second-order GVD parameter, respectively, γ_k is the nonlinearity coefficient and v_{gk} is the group velocity.

In order to simplify the analysis and focus our attention on the effect of XPM induced inter channel crosstalk, we neglect

the interaction between SPM and XPM and assume that these two act independently. We will assume that the probe signal is operated in continuous wave (CW), whereas the pump signal is modulated with a sinusoidal wave at a frequency Ω .

Using the substitutions $T=t-z/v_k$ and $\alpha_k = \alpha$

$$A_k(t, z) = E_k(T, z) \exp(-\alpha z/2) \quad (2)$$

we have,

$$\frac{\partial E_k(T, z)}{\partial z} = -\frac{i\beta_{2k}}{2} \frac{\partial^2 E_k(T, z)}{\partial T^2} + \frac{\beta_{3k}}{6} \frac{\partial^3 E_k(T, z)}{\partial T^3} + i\gamma_k 2P_k(T - d_{jk}, 0) \cdot \exp(-\alpha z) E_k(T, z) \quad (3)$$

where $d_{jk} \equiv (1/v_j) - (1/v_k)$ is the relative walk-off between the probe and the pump. Using a linear approximation, the walk-off d_{jk} can be expressed as $d_{jk} = D\Delta\lambda_{jk}$, where $D = -(2\pi/\lambda^2)\beta_{2k}$ is the dispersion coefficient and second order GVD, $\beta_3 = \frac{\lambda^3}{2\pi^2 c^2} \left[D(\lambda) + \frac{\lambda}{2} \cdot \text{dispersion slope} \right]$, $\Delta\lambda_{jk}$ and λ are the wavelength spacing and the average wavelength between the probe and pump, respectively, and c is the light velocity. Here a linear approximation is used for d_{jk} for simplicity. In general, dispersion and non-linearity act together along the fiber. However, in an infinitesimal fiber section dz , it can be assumed that dispersive and non-linear effects act independently. Now, substituting $E_k = |E_k| \exp[i\Phi_k(T, z)]$ in equation (3) gives,

$$d\phi_k(T, z') = i\gamma_k 2P_j(T + d_{jk}z', 0) \exp(-\alpha z') dz \quad (4)$$

The Fourier transformation of this phase variation gives

$$d\phi_k(\Omega, z') = i\gamma_k 2P_j(\Omega, 0) e^{(-\alpha + i\Omega d_{jk})z'} dz \quad (5)$$

Neglecting the intensity fluctuations of the probe channel, this phase change corresponds to change in electrical field,

$$|E_k| \exp[i d\phi_k(T, z')] \approx |E_k| [1 + i d\Phi(T, z')] \quad (5a)$$

Or in Fourier domain (5a) can be written as,

$$|E_k| \exp[i d\phi_k(\Omega, z')] \approx |E_k| [1 + i d\Phi(\Omega, z')] \quad (5b)$$

Due to chromatic dispersion of the fiber, this phase variation generated at $z = z'$ is converted into an amplitude variation at the end of the fiber $z = L$. Taking into account only the dispersion and source term of the phase perturbation at $z = z'$, the Fourier transform of equation (3) becomes

$$\frac{\partial E_k(\Omega, z)}{\partial z} = -\frac{i\beta_{2k}\Omega^2}{2} E_k(\Omega, z) - \frac{\beta_{3k}\Omega^3}{6} E_k(\Omega, z) + |E_k| [1 + i d\Phi(\Omega, z')] \delta(z - z') \quad (6)$$

where the Kronecker delta $\delta(z - z')$ is introduced to take into account the fact the source term exists only in the infinitesimal fiber section at $z = z'$. Here we only consider to study the effect of second order GVD since first order GVD are now totally compensated. So we eliminated β_{2k} from equation (6). Therefore at $z = L$ the probe field becomes

$$E_k(\Omega, L) = E_k + i d\Phi_k(\Omega, z') E_k \exp \left[\left(-\frac{\beta_{3k}\Omega^3}{6} \right) (L - z') \right] \quad (7)$$

The optical power variation caused by non-linear phase modulation created in the short section dz at $z = z'$ is thus

$$\Delta a_{jk}(\Omega, z', L) = |E_k(\Omega, L)|^2 - E_k^2 = -2E_k^2 d\Phi_k(\Omega, z') \sin \left[\left(\frac{i\beta_{3k}\Omega^3}{6} \right) (L - z') \right] \quad (8)$$

where linearization has been made considering that $d\phi_k$ is infinitesimal.

Using $E_k(T, z) = A_k(T + z/v_k, z) \exp(\alpha z/2)$ and equation (5) and integrating all self phase modulation and cross phase modulation contributions along the fiber, gives the total intensity fluctuations at the end of the fiber.

$$\Delta s_{jk}(\Omega, L) = -4\gamma_k E_k^2 \exp(-(\alpha - i\Omega/v_k)L) \int_0^L P_j(\Omega, 0) \cdot \sin \left[\left(i\beta_{3k}\Omega^3/6 \right) (L - z') \right] \exp(-(\alpha - i\Omega d_{jk})z') dz' \exp(-(\alpha - i\Omega d_{jk})z') dz' \quad (9)$$

Where $P_k = |A_k|^2$ and $\Delta s_{jk}(\Omega, L) = \Delta a_{jk}(\Omega, L) e^{-\alpha L}$ represents fluctuations of A_k . Which on integration gives

$$\Delta s_{jk}(\Omega, L) = 2\gamma_k P_k(L) e^{i\Omega L/v_k} \left\{ \frac{\exp(\beta_{3k}\Omega^3 L/6) - \exp(-\alpha + i\Omega d_{jk})L}{i(\alpha - i\Omega d_{jk} - \beta_{3k}\Omega^3/6)} \right\} - 2\gamma_k P_k(L) e^{i\Omega L/v_k} \left\{ \frac{\exp(-\beta_{3k}\Omega^3 L/6) - \exp(-\alpha + i\Omega d_{jk})L}{i(\alpha - i\Omega d_{jk} - \beta_{3k}\Omega^3/6)} \right\} \quad (10)$$

where $P_k(0)$ and $P_k(L)$ are probe optical power at input and output of the fiber. Under the assumption that $\exp(-\alpha z) \ll 1$ gives a simpler frequency domain description of the intensity fluctuations in the probe channel by the pump channel.

$$\Delta s_{jk}(\Omega, L) = 4\gamma_k P_k(L) P_j(\Omega, 0) \cdot \exp(i\Omega L/v_k) \cdot \frac{\sin(\beta_{3k}\Omega^3 L/6)}{\alpha - i\Omega d_{jk}} \quad (11)$$

Equation (11) can be generalized to analyze multispan optically amplified systems, where the total intensity fluctuation at the receiver is the accumulation of XPM contributions created by each fiber span. For a system having a total of N amplified fiber spans, the XPM created in the m th span produces an intensity modulation $\Delta s_{jk}^{(m)}(\Omega, L_N)$ at the end of the system. Even through the phase modulation creation depends only on the pump power and the walk-off within the m th span, the phase-to-intensity conversion depends on the accumulated dispersion of the fibers from the m th to the N th spans. There fore, we have

$$\Delta s_{jk}^{(m)}(\Omega, L_N) = 4\gamma_k P_k(L_N) P_j^{(m)}(\Omega, 0) \cdot \exp \left[i\Omega \sum_{n=1}^{m-1} d_{jk}^{(n)} L^{(n)} \right] \cdot \frac{\sin \left[\left(\Omega^3 \sum_{n=m}^N \beta_{3k}^{(n)} L^{(n)} \right) / 6 \right]}{\alpha - i\Omega d_{jk}^{(m)}} \cdot \exp(-i\Omega L_N/v_k) \quad (12)$$

where $L_N = \sum_{n=1}^N L^{(n)}$ is the total fiber length in the system,

$L^{(m)}$ and β_3^m are the fiber length and the dispersion of the m th span [where $L^{(0)} = 0$], $P_k^{(m)}(\Omega, 0)$ is the pump signal input power spectrum in the m th span and $d_{jk}^{(m)}$ is the relative walk-off between two channels in the m^{th} span, [where $d_{jk}^{(0)} = 0$]. To generalize (11) to (12), the term $\sin(\beta_{3k}\Omega^3 L/6)$ in (11) is replaced by $\sin \left[\left(\Omega^3 \sum_{n=m}^N \beta_{3k}^{(n)} L^{(n)} \right) / 6 \right]$ in (12) to take into account the linear accumulation of dispersion. The phase difference between the pump and the probe waves at the input of the m^{th} span is different from that at the input of the first

span. The walk-off dependent term $\exp \left[i\Omega \sum_{n=1}^{m-1} d_{jk}^{(n)} L^{(n)} \right]$ in (12) takes into account the walk-off between the probe and the pump channels before they both enter into the m^{th} fiber span.

Finally, contributions from all fiber spans add up and therefore, the intensity fluctuation induced by the XPM of the whole system can be expressed as

$$\Delta s_{jk}(\Omega, L_N) = \sum_{m=1}^N \Delta s_k^{(m)}(\Omega, L_N) \quad (13)$$

In the time domain, the probe output optical power with XPM induced crosstalk is,

$$P_{jk}(t, L_N) = p_k(L_N) + \Delta s_{jk}(t, L_N) \quad (14)$$

$\Delta s_{jk}(t, L_N)$ is the inverse Fourier transform of $\Delta s_{jk}(\Omega, L_N)$ and $p_k(L_N)$ is the probe output without XPM. After the square law detection of photo diode, the electrical power spectral density is the Fourier transform of the correlation of the time domain optical intensity waveform. Therefore,

$$P_k(\Omega, L_N) = \eta^2 \left[P_k^2(L_N) \delta(\Omega) + |\Delta s_{jk}(\Omega, L_N)|^2 \right] \quad (15)$$

where η is the photodiode responsivity.

For $\Omega > 0$, the XPM induced electrical power spectral density in the probe channel, normalized to its power level without this effect can be expressed as

$$\Delta P_{jk}(\Omega, L_N) = \frac{\eta^2 |\Delta s_{jk}(\Omega, L_N)|^2}{\eta^2 P_k^2(L_N)} \quad (16)$$

$\Delta P_{jk}(\Omega, L_N)$ is defined as the normalized XPM power transfer function.

$$\Delta P_{jk}(\Omega, L_N) = \left\{ \frac{4 \gamma_k P_k^j(\Omega, 0) \cdot \exp \left[i\Omega \sum_{n=1}^{i-1} d_{jk}^{(n)} L^{(n)} \right] \cdot \sin \left(\Omega^3 \sum_{n=i}^N \beta_{3k}^{(n)} L^{(n)} / 6 \right)}{\alpha - i\Omega d_{jk}^i} \right\}^2 \quad (17)$$

which is the final expression. It is worth mentioning here that in the derivation of (17), we have neglected the intensity fluctuation of the probe signal before it reaches the end of the system.

IV. RESULTS AND DISCUSSION

Following the derived theoretical expressions in the preceding section, numerical solutions have been carried out by varying different parameters. Here we have used 3-types of fibers, such as- SSMF, DSF and LEAF, and compared their XPM induced crosstalk performance in presence of second order GVD only. Though the analytical model is developed for multi-channel and multi-span system, most results are depicted for pump-probe configuration. Different system parameters of 3-types of fibers used in simulation are shown in Table 1.

To compare the amount of crosstalk among SSMF, DSF and LEAF, we have plotted crosstalk versus modulation frequency (span length 100 km) in presence of second order GVD only in Fig. 2 using different channel spacing. The results show that XPM crosstalk penalty due to second order GVD is 43 dB more for DSF with respect to SSMF and it is about 2.0 dB more than that of LEAF at 10 GHz modulation frequency and 0.8 nm channel spacing.

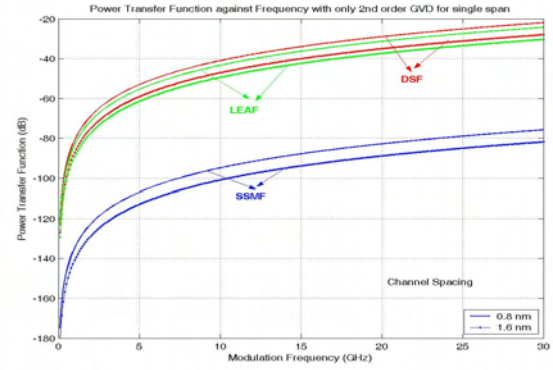


Fig. 2: Plots of crosstalk vs. modulation frequency with second order GVD for single span for three types of fiber.

Although we observed that the effect of second order GVD on the system performance is not significant for single span system. But in practical long haul WDM system, the effect of second order becomes significant. This is depicted in Fig. 3, where we have determined the effect of second order GVD only for 0.8 nm channel spacing for various span lengths of SSMF. From the figure it is found that at 10 GHz modulation frequency the crosstalk is -105 dB, -78 dB and -65 dB for fiber length of 100 km, 1000 km and 10,000 km respectively.

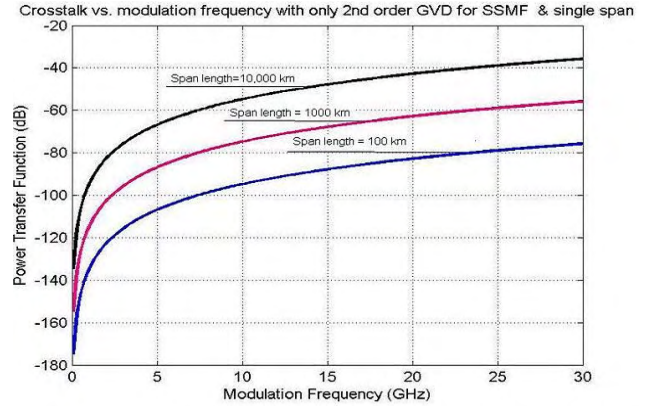


Fig. 3: Crosstalk vs. modulation frequency for various span lengths in presence of second order GVD only.

Now, the XPM effect in a 2-span optical WDM system is observed in presence of second order GVD only and it shown in Fig.4. From Fig. 4, we have noted that the shape of XPM frequency response is dependent on the channel spacing.

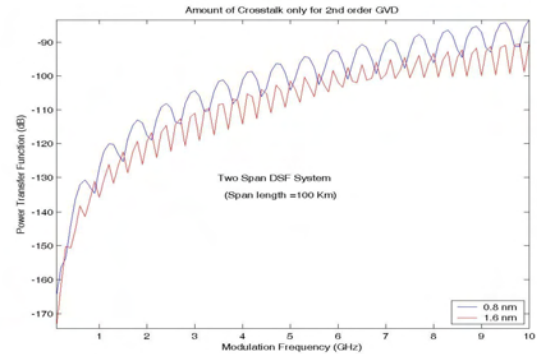


Fig. 4: plots of crosstalk vs. modulation frequency with only second order GVD for two span using DSF.

Here, ripples are observed in the plots (XPM induced crosstalk vs. modulation frequency). These ripples are due to interference between XPM-induced crosstalk which is created in previous fiber span. It can be shown from equation (17) that the notches in the spectrum are located at frequencies which satisfy approximately the relation,

$$1 + e^{i\Omega d_{jk} L_1} = 0 \quad (18)$$

and, therefore, the frequency difference between adjacent notches of a spectrum is $\Delta f = 1/(d_{jk} L_1)$, where L_1 is the fiber length of the first span

Fig. 5 shows the impact of XPM crosstalk vs. channel number for SSMF. From the results, it is found that as the channel number increases XPM crosstalk also increases for a fixed channel spacing & modulation frequency. For example, from Fig. 5 shows that in SSMF (channel spacing 0.8 nm and modulation frequency: 5 MHz) amount of crosstalk is -53 dB and -62 dB for 8 channels and 2 channels respectively. So, we can conclude that crosstalk is directly proportional to the number of channels.

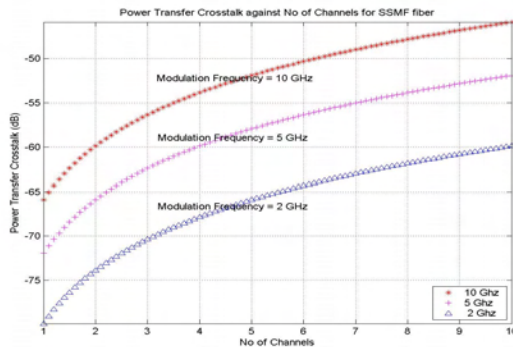


Fig. 5: Plots of crosstalk vs. number of channel for SSMF Fiber with various modulation frequencies.

TABLE 1: SIMULATION PARAMETERS FOR DIFFERENT FIBERS

Parameter (unit)	SSMF	DSF	LEAF
Probe wavelength (nm)	1559	1559	1559
Channel spacing (nm)	0.8, 1.6	0.8, 1.6	0.8, 1.6
Zero dispersion slope (ps/nm ² /nm)	0.095	0.075	0.084
Effective Area (m ²)	8.0X10 ⁻¹¹	5.5X10 ⁻¹¹	7.2X10 ⁻¹¹
Attenuation parameter (dB/km)	0.25	0.25	0.25
Dispersion Parameter (ps/nm/km)	17	3.5	4.5
Second order GVD (ps ³ /nm)	0.192295	0.1995	0.1957

V. CONCLUSION

In this paper, a detailed analysis is carried out for optical crosstalk due to XPM in presence of second order GVD in a WDM fiber-optic transmission system. Results are evaluated and compared for commercially available SSMF, DSF and LEAF. It is clear that the crosstalk level due to XPM and GVD increases with increasing number of WDM channels, modulating frequency and channel spacing. We observed that XPM has more impact on DSF fiber than that of SSMF and LEAF. Thus in long haul and high bit rate WDM fiber-optic system second order GVD plays a crucial role in limiting the system performance significantly.

REFERENCES

- [1] G. P. Agrawal "Nonlinear Fiber Optics." Academic Press, New York, 2002
- [2] M. Nishimura. "Optical fibers and fiber dispersion compensators for high-speed optical communication", Journal of Optical Fiber Communication Report 2, Editor: Hans-Georg Weber, Masataka Nakazawa, pp. 115–139, 2005.
- [3] E. J. Gualda, L. C. Go´mez-Pavó And J. P. Torres, "Compensation of third-order dispersion in a 100 Gb/s single channel system with in-line fiber Bragg gratings", Journal of Modern Optics, vol. 52, no. 9, pp. 1197–1206, 2005.
- [4] Y. M. Karfaa, M. Ismail, F. M. Abbou, S. shaari, and S. P. Majumder. "Effects of cross-phase modulation crosstalk in WDM networks on received power and number of channels for various fiber types", Proc. of Asia Pacific Conference on Applied Electromagnetic (APCAE 2007), Malaysia, vol. 1, pp 22-27, 2007.
- [5] H. A. Abdul-Rashid, F. M. Abbou, H. T. Chuah, M. B. Tayahi, M. T. Al-Qdah, and S. Lanka. "System performance optimization in SCM-WDM passive optical networks in the presence of XPM and GVD", IEEE Communications Letters, vol. 10, 2006.
- [6] Y. M Karfaa, M. Ismail, F. M. Abbou, S. Shaari, and S. P. Majumder, "Channel spacing effects on XPM crosstalk in WDM networks for various fiber types", Proc. of IEEE 2008 6th National Conference on Telecommunication Technologies and Photonics, Malaysia, 2008.
- [7] H. A. A. Rashid, F. M. Abbou, M. T. Al-Qdah, H. T. Chuah and M. B. Tayahi, "Semi-analytical performance analysis of an SCM-WDM PON in the presence of XPM and GVD", IEEE International Conference on Telecommunications, Malaysia, pp.35-38, 2007.
- [8] R. Hui, K. R. Demarest and C. T. Allen. "Cross-phase modulation in multispan WDM optical fiber systems", Journal of Lightwave Technology, vol. 17, no. 6, pp. 1018-1026, 1999.